Т3

$$p(x) = \begin{cases} e^{-\frac{x}{\theta}} \cdot \frac{1}{\theta}, & x \geq 0 \\ 0, & x < 0 \end{cases}, \quad \theta > 0$$

Объем выборки $n = 3$

Оценки $\tilde{\theta}_1 = \bar{x}$, $\tilde{\theta}_3 = x_{(2)}$

$$F(x) = \begin{cases} 1 - e^{-\frac{x}{\theta}}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$M\xi = \int_0^\infty x \cdot \frac{1}{\theta} e^{-\frac{x}{\theta}} dx = \left| \begin{array}{l} t = \frac{x}{\theta} \\ x = \theta t \\ dx = \theta dt \end{array} \right| = \theta \int_0^\infty t e^{-t} dt = \theta$$

$$M\xi^2 = \int_0^\infty \frac{x^2}{\theta} \cdot e^{-\frac{x}{\theta}} dx = \theta^2 \int_0^\infty t^2 e^{-t} dt = 2\theta^2$$

$$D\xi = M\xi^2 - (M\xi)^2 = 2\theta^2 - \theta^2 = \theta^2$$

а) ① $\tilde{\theta}_1 = \frac{1}{n} \sum_{i=1}^n x_i$

$$M\tilde{\theta}_1 = M\left(\frac{1}{n} \sum_{i=1}^n x_i\right) = \frac{1}{n} \sum_{i=1}^n M x_i = \frac{1}{n} \cdot n\theta = \theta$$ - не смещенная

② $\tilde{\theta}_2 = x_{(2)}$

$$\boxed{f_{(k)} \sim \sum_{i=k}^n C_n^i F^i(t)(1-F(t))^{n-i}}$$

$$F_{x_{(2)}} = \sum_{i=2}^3 C_3^i F^i(t)(1-F(t))^{3-i} = C_3^2 F^2(t)(1-F(t))^1 + C_3^3 F^3(t)(1-F(t))^0 =$$

$$= 3F^2(t)(1-F(t)) + F^3(t)$$

$$F_{X_{(2)}} = 3\left(1-e^{-\frac{t}{\theta}}\right)^2 - 2\left(1-e^{-\frac{t}{\theta}}\right)^3$$

=> плотность $\quad p_{X_{(2)}} = 3\cdot 2F(t)F'(t) - 2\cdot 3F^2(t)F'(t) =$

$$= 6F(t)F'(t) - 6F^2(t)F'(t) = 6F'(t)F(t)(1-F(t))$$

$$F'(t) = \frac{1}{\theta} e^{-y/\theta}$$

$$p_{X_2} = 6\frac{1}{\theta} e^{-\frac{y}{\theta}}\left(1-e^{-\frac{y}{\theta}}\right)\left(1-\left(1-e^{-\frac{y}{\theta}}\right)\right) =$$

$$= \frac{6}{\theta} e^{-\frac{2y}{\theta}}\left(1-e^{-y/\theta}\right)$$

$$p_{X_{(2)}} = \begin{cases} \frac{6}{\theta}\left(e^{-\frac{2y}{\theta}} - e^{-\frac{3y}{\theta}}\right), & y \geq 0 \\ 0, & y < 0 \end{cases}$$

$$M\tilde{\theta}_2 = \int_0^\infty y \cdot \frac{6}{\theta}\left(e^{-\frac{2y}{\theta}} - e^{-\frac{3y}{\theta}}\right)dy =$$

смещенная

$$= \frac{6}{\theta}\int_0^\infty y e^{-2y/\theta}dy - \frac{6}{\theta}\int_0^\infty y e^{-3y/\theta}dy = \frac{6}{\theta}\left(\frac{\theta^2}{4} - \frac{\theta^2}{9}\right) = \frac{5\theta}{6}$$

$\tilde{\theta}_2' = \frac{6}{5}\tilde{\theta}_2$ – несмещенная

б) ① $D\tilde{\theta}_1 = \frac{1}{n} D\xi = \frac{\theta^2}{3}$

② $M\tilde{\theta}_2^2 = \int_0^\infty \frac{6}{\theta} y^2 \left(e^{-\frac{2y}{\theta}} - e^{-\frac{3y}{\theta}}\right) dy =$

$= \frac{6}{\theta}\left(\int_0^\infty y^2 e^{-\frac{2y}{\theta}} dy - \int_0^\infty y^2 e^{-\frac{3y}{\theta}} dy\right) = \frac{6}{\theta}\left(\frac{\theta^3}{4} - \frac{2\theta^3}{27}\right) = \frac{19}{18}\theta^2$

$D\tilde{\theta}_2 = M\tilde{\theta}_2^2 - (M\tilde{\theta}_2)^2 = \frac{19}{18}\theta^2 - \frac{25}{36}\theta^2 = \frac{13}{36}\theta^2$

$\Rightarrow \tilde{\theta}_2' = \frac{6}{5}\tilde{\theta}_2 \Rightarrow D\tilde{\theta}_2' = \left(\frac{6}{5}\right)^2 \tilde{\theta}_2 = \frac{36}{25}\cdot\frac{13}{36}\theta^2 = \frac{13}{25}\theta^2$

Получаем, что $D\tilde{\theta}_1 < D\tilde{\theta}_2'$

Если $\forall n$

$D\tilde{\theta}_1 = \frac{\theta^2}{n}$ $\quad D\tilde{\theta}_2' = \frac{n^2(n-1)^2}{(2n-1)^2} D\tilde{\theta}_2 = \frac{2n^2-2n+1}{4n^2-4n+1}\theta^2$

Тогда $\forall n > 1$ $\quad D\tilde{\theta}_1 < D\tilde{\theta}_2'$

c) Неравенство Крамера-Рао:

Если модель регулярная и стат. $\hat{g}(\vec{x_n})$ явл. регул. оценкой $g(\theta)$, тогда вып.

$$D\hat{g}(\vec{x_n}) \geq \frac{(g'(\theta))^2}{nI(\theta)} \quad \forall \theta \in \Theta$$

① Регулярность модели

1. $p(x) = \begin{cases} \frac{1}{\theta} \cdot e^{-\frac{x}{\theta}}, & x \geq 0 \\ 0, & x < 0 \end{cases}$ $\quad \Theta = (0, \infty)$

непр. дифф по θ на Θ

2. $\int_0^\infty \left(\frac{1}{\theta} \cdot e^{-\frac{x}{\theta}} \cdot \frac{x}{\theta^2} - \frac{1}{\theta^2} e^{-\frac{x}{\theta}}\right) dx = \frac{1}{\theta} + \frac{1}{\theta}(0-1) = 0 =$

$= \frac{\partial}{\partial \theta}\left(\int_0^\infty \frac{1}{\theta} e^{-\frac{x}{\theta}} dx\right)$ $\quad \int_0^\infty \frac{\partial p(x,\theta)}{\partial \theta} dx = 0$

3. $\ln\left(\frac{1}{\theta} \cdot e^{-x/\theta}\right) = -\frac{x}{\theta} - \ln\theta$

$\left(\frac{\partial \ln p}{\partial \theta}\right)^2 = \left(\frac{x}{\theta^2} - \frac{1}{\theta}\right)^2$

$I(\theta) = M\left[\left(\frac{\partial \ln p}{\partial \theta}\right)^2\right] = \int_0^\infty \left(\frac{x^2}{\theta^4} - 2\frac{x}{\theta^3} + \frac{1}{\theta^2}\right)\frac{1}{\theta} e^{-\frac{x}{\theta}} dx =$

$= \frac{1}{\theta^2}\left[\int_0^\infty t^2 e^{-t} dt - 2\int_0^\infty t e^{-t} dt + \int_0^\infty e^{-t} dt\right] =$

$= \frac{1}{\theta^2}(2 \cdot 2 + 1) = \frac{1}{\theta^2}$ - непрер., на Θ $\frac{1}{\theta^2} > 0$

$\Rightarrow$ Модель регулярна

② Регулярность оценки

1. $\tilde{\theta}_1$ - несмещ. оценка θ, $D\tilde{\theta}_1 = \frac{\theta^2}{n}$ - огр. на $\forall$ компакте из Θ по θ $\Rightarrow$ $\tilde{\theta}_1$ - регул.

2. $\tilde{\theta}_2'$ - несмещ. оценка θ, $D\tilde{\theta}_1' = \frac{13}{25}\theta^2$ - огр. на $\forall$ компакте из Θ по θ $\Rightarrow$ $\tilde{\theta}_2'$ - регулярна

Нер-во Крамера-Рао использовать можем

1) Модель регул., $\tilde{\theta}_1$ - регул., $g(\theta) = \theta$ - дифф $\Rightarrow$

$$\forall \theta \in \Theta \quad D\tilde{\theta}_1 \geq \frac{(g'(\theta))^2}{n I(\theta)} = \frac{1}{3 \cdot 1/\theta^2} = \frac{\theta^2}{3}$$

$\Rightarrow$ $\tilde{\theta}_1$ - эффективная, т.к. $D\tilde{\theta}_1 = \frac{\theta^2}{3}$ по дост. усл.

2) Модель регул., $\tilde{\theta}_2'$ - регул., $g(\theta) = \theta$ - дифф $\Rightarrow$

$\forall \theta \in \Theta \quad D\tilde{\theta}_2' \geq \dfrac{(g'(\theta))^2}{n\, I(\theta)} = \dfrac{\theta^2}{3}$

Т.к. $\dfrac{13}{25}\theta^2 \neq \dfrac{\theta^2}{3}$, оценка $\tilde{\theta}_2'$ не явл.

эффективной, нер-во ничего не говорит

Также, если эффективная оценка $\exists$, то она ! $\Rightarrow$ если $\tilde{\theta}_2' \neq \tilde{\theta}_3$, а $\tilde{\theta}_3$ - эфф. $\Rightarrow$

$\Rightarrow \tilde{\theta}_2'$ не эфф.

Т4 $\xi \sim \rho(x) = p\{(-1,1)\setminus\{0\}\} + p_1\{0\} + p_1\{2\}$

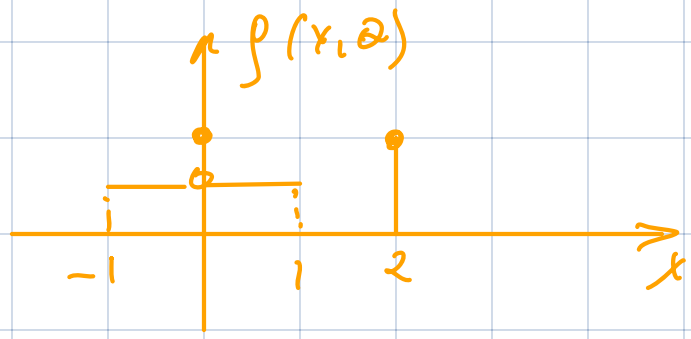
$\int_{-\infty}^{\infty} \rho(x)\,dx = 1 \Rightarrow \int_{-1}^{1} p\,dx + 2p_1 = 2p + 2p_1 = 1 \Rightarrow$

$\Rightarrow p_1 = \frac{1}{2} - p$

$p = \theta \quad ; \quad p_1 = \frac{1}{2} - \theta$

$\rho(x,\theta) = \theta\{(-1,1)\setminus\{0\}\} + \left(\frac{1}{2} - \theta\right)\{0\} + \left(\frac{1}{2} - \theta\right)\{2\}$

$\theta \in \left(0; \frac{1}{2}\right)$



$\vec{X}_n$ - выборка

1) О.М.М.

а) $\alpha_1 = \int\limits_{-\infty}^{\infty} x\rho(x,\theta)dx = \int\limits_{-1}^{1} \theta \cdot x\,dx + 0 + 2\left(\frac{1}{2} - \theta\right) = 1 - 2\theta$

$\alpha_1(\theta) = \tilde{\alpha}_1 = \bar{x} = \frac{1}{n}\sum\limits_{i=1}^{n} x_i \qquad M\xi = \alpha_1 = 1 - 2\theta$

$1 - 2\theta = \bar{x} \Rightarrow \tilde{\theta}_1 = \frac{1-\bar{x}}{2}$

в) $M\tilde{\theta}_1 = \frac{1}{2} - \frac{1}{2}M\xi = \frac{1}{2} - \frac{1}{2} + \frac{1}{2}\cdot 2\theta = \theta \Rightarrow$ несмещенная

$M\xi^2 = \int\limits_{-1}^{1} \theta x^2 dx + 4\left(\frac{1}{2} - \theta\right) = \frac{2\theta}{3} + 2 - 4\theta =$

$= 2 - \frac{10}{3}\theta$

$D\xi = M\xi^2 - (M\xi)^2 = 2 - \frac{10}{3}\theta - (1-2\theta)^2 =$

$= 1 + \frac{2}{3}\theta - 4\theta^2$

$D(\tilde{\theta}_1) = D\left(\frac{1}{2} - \frac{\bar{x}}{2}\right) = \frac{1}{4}D\bar{x}$

$$D\bar{x} = \frac{1}{n^2} \cdot n D\xi = \frac{D\xi}{n}$$

$$D(\tilde{\theta}_1) = \frac{D\xi}{4n} = \frac{1 + \frac{2}{3}\theta - 4\theta^2}{4n} \xrightarrow[n \to \infty]{} 0$$

$\Rightarrow \tilde{\theta}_1$ состоят. по дост. условию

e) Проверим регул. модели

1) $\int_{-\infty}^{\infty} \frac{\partial}{\partial \theta} f(x, \theta)\, dx = 0$

$$\int_{-1}^{1} 1\, dx + (-1) + (-1) = 2 - 2 = 0$$

2) $p(x, \theta) = \theta$ непр. дифф.

$p_k(\theta) = \frac{1}{2} - \theta$ по θ на $\ominus$

3) $$I(\theta) = \int_{-1}^{1} \left(\frac{\partial \ln p(x, \theta)}{\partial \theta} \right)^2 p(x, \theta)\, dx +$$

$$+ \sum_k \left(\frac{\partial \ln p_k(\theta)}{\partial \theta} \right)^2 p_k(\theta) =$$

$$= \int_{-1}^{1} \frac{1}{\theta}\, dx + 2\left(\frac{1}{2} - \theta\right) \left(\frac{\partial \ln(\frac{1}{2} - \theta)}{\partial \theta} \right)^2 = \frac{2}{\theta} + 2 \cdot \frac{1}{(\frac{1}{2} - \theta)} =$$

$= \dfrac{2}{\theta(1-2\theta)} > 0$ и непрер. на $(0; \frac{1}{2})$

$\Rightarrow$ модель регулярна

Проверим регул. оценки

$D\tilde{\theta}_1 = \dfrac{1 + \frac{2}{3}\theta - 4\theta^2}{4n}$ - огр. на $\forall$ компакте из $(0; \frac{1}{2})$

$\Rightarrow$ рег. по рост. усл.

$\Rightarrow$ Можно прим. Крамера-Рао

$$D\tilde{\theta}_1 \geq \frac{1}{n \dfrac{2}{\theta(1-2\theta)}} = \frac{\theta(1-2\theta)}{2n}$$

$$D\tilde{\theta}_1 = \frac{1 + \frac{2}{3}\theta - 4\theta^2}{4n}$$

Получаем $\frac{1}{2} + \frac{\theta}{3} > \theta$

$D\tilde{\theta}_1 > \dfrac{1}{n\, I(\theta)}$ $\Rightarrow$ ничего не знаем.

2) О.М.П.

1) $L(\theta) = \prod_{i=1}^{n} p(x_i, \theta)$ – ф-ия правдоподобия

Пусть в выборке $\vec{x_n}$ m раз встретилось $\{0\}$ или $\{2\}$

$$L(\theta) = \theta^{n-m}\left(\frac{1}{2}-\theta\right)^m$$

$$\ln L(\theta) = (n-m)\ln\theta + m\ln\left(\frac{1}{2}-\theta\right)$$

$$\frac{\partial}{\partial\theta}\ln L(\theta) = \frac{n-m}{\theta} - \frac{m}{\frac{1}{2}-\theta} = \frac{(n-m)\left(\frac{1}{2}-\theta\right)-m\theta}{\theta\left(\frac{1}{2}-\theta\right)} =$$

$$= 0$$

$$\frac{n}{2} - n\theta - \frac{m}{2} = 0 \qquad \theta = \frac{n-m}{2n} = \frac{1}{2} - \frac{1}{2}\nu$$

$$\frac{\partial^2 \ln L(\theta)}{\partial\theta^2} = \frac{m-n}{\theta^2} + \frac{m}{\left(\frac{1}{2}-\theta\right)^2}(-1)^3 =$$

$$= -\frac{n-m}{\theta^2} - \frac{m}{\left(\frac{1}{2}-\theta\right)^2} < 0 \quad \text{при } \theta \in \left(0, \frac{1}{2}\right)$$

2) $M\tilde{\theta}_2 = \frac{1}{2} - \frac{1}{2}M\nu = \frac{1}{2} - \frac{1}{2}p = \frac{1}{2} - \frac{1}{2}(1-2\theta) =$

$= \frac{1}{2} - \frac{1}{2} + \theta = \theta \Rightarrow$

несмещенная

3) $D\tilde{\theta}_2 = D\left(\frac{1}{2} - \frac{1}{2}\nu\right) = \frac{1}{4}D\nu = \frac{1}{4}\frac{p(1-p)}{n} =$

$= \frac{1}{4}\left(\frac{2\left(\frac{1}{2}-\theta\right)(2\theta)}{n}\right) = \frac{\theta\left(\frac{1}{2}-\theta\right)}{n} = \frac{\theta(1-2\theta)}{2n}$

$\xrightarrow[n\to\infty]{} 0 \Rightarrow$ оценка состоят

$D\tilde{\theta}_2$ - огр. на $\forall$ комп. из $(0;\frac{1}{2}) \Rightarrow$ рег.

Модель регулярна

$D\tilde{\theta}_2 = \frac{\theta(1-2\theta)}{2n} \geq \frac{\theta(1-2\theta)}{2n} \Rightarrow$

$\tilde{\theta}_2$ эффективная $\Rightarrow$ $\tilde{\theta}_1$ - не эффект.